Minimal-Shot Autonomous Driving

SoTA Commission I Submission

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GitHub branch: `CoRL-2027`

Goal: build an autonomous-driving prototype that can face unfamiliar long-tail scenarios by reasoning from geometry, not memorizing a mapped training world.

Motivation

Autonomy that depends on collecting every rare case will always lag reality.

Minimal-shot autonomy needs a different capability:

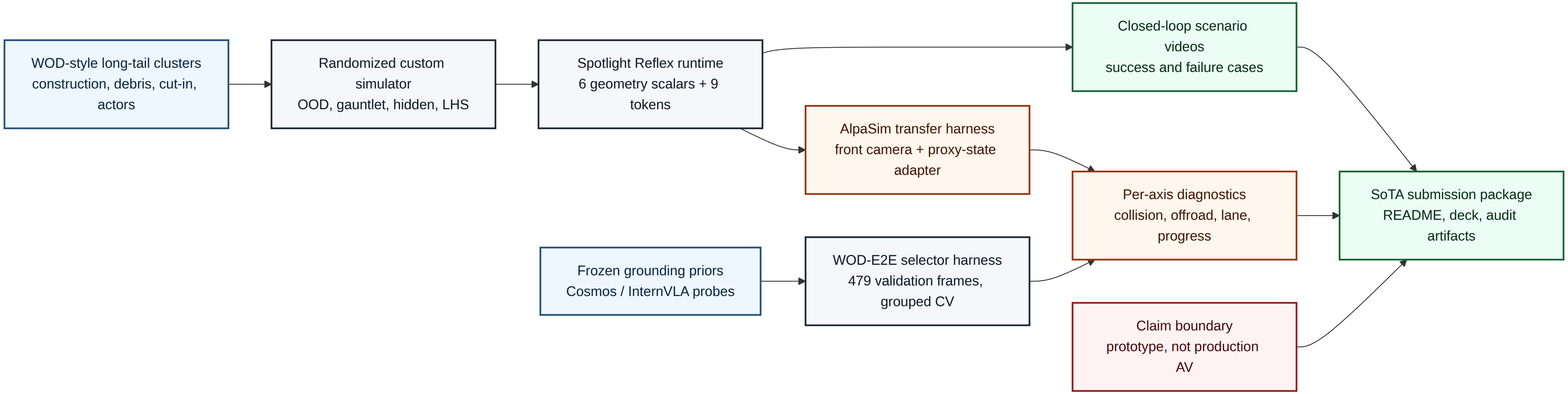
Standard AV scaling	Minimal-shot target
train on many mapped cases	reason from first principles
optimize average-case driving	survive long-tail scenarios
hide failures inside aggregate metrics	expose failures by scenario and axis
expensive data dependence	reusable simulator and audit harness

This project asks whether a small, auditable driving system can handle rare geometry shifts without fine-tuning on an AV dataset.

What Was Built For The Challenge

SoTA requirement	Delivered artifact
simulation environment	custom 2D long-tail AV simulator
randomized scenarios	WOD-style clusters, OOD generator, gauntlet, LHS sweeps
autonomous system demo	Spotlight Reflex token policy + rollout GIFs
realistic constraints	small scalar runtime, deterministic controller, no heavy online model
WOD-E2E pathway	selector harness over real WOD-E2E validation frames
[illegible]	reproducible scripts, audit reports, presentation, paper

Whole System Map



Mermaid-authored system schema. The architecture has three connected layers:

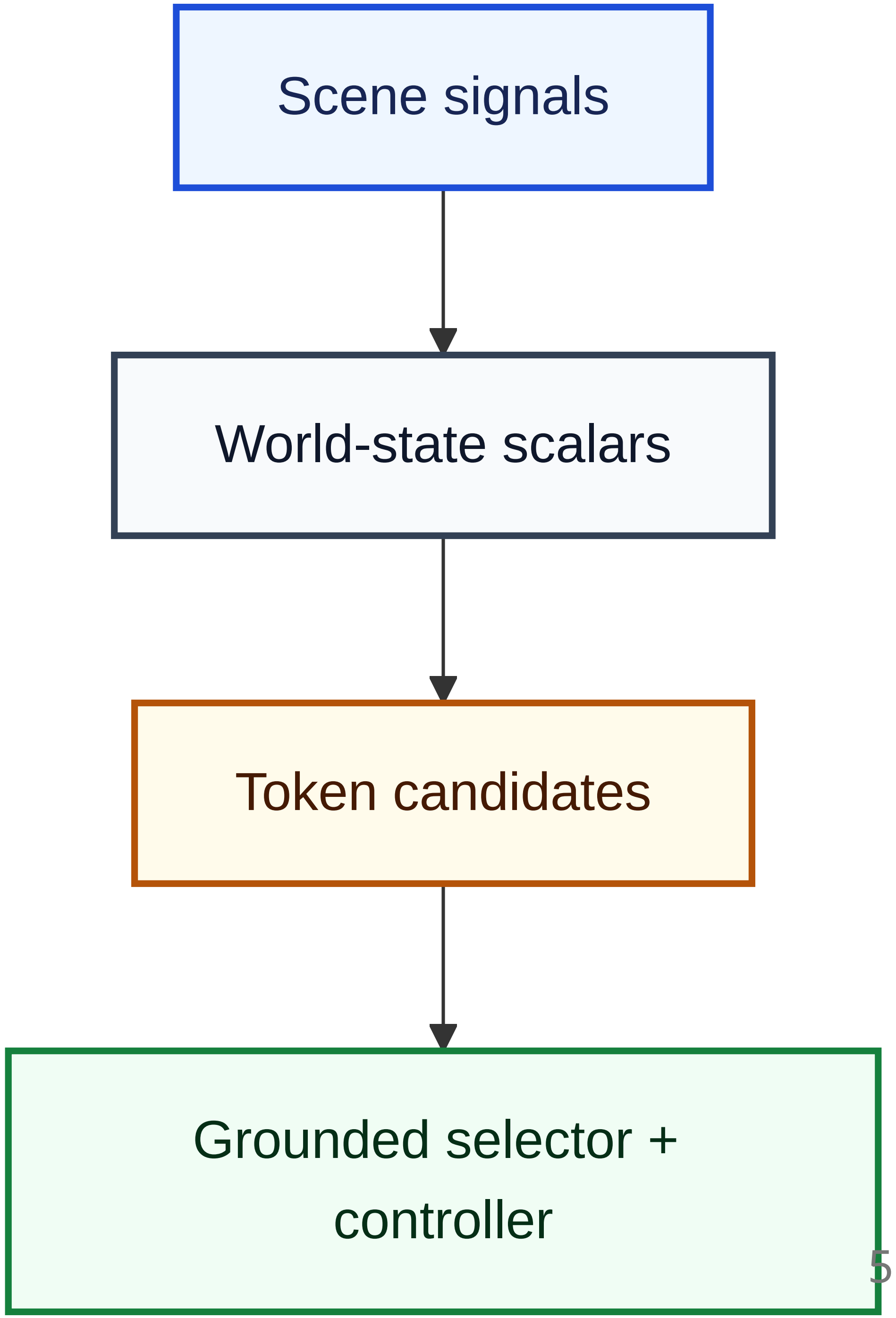
1. custom simulator for controlled minimal-shot experiments
2. WOD-E2E selector harness for real-data grounding probes
3. AlpaSim transfer harness for sensor-realistic external validation

Runtime Architecture

Spotlight Reflex converts visible geometry into a small token decision:

- six world-state scalars describe route blockage and clearances
- nine ManeuverTokens define candidate actions
- trust-region rules score candidate futures
- the controller executes the selected token

The runtime does not use scene IDs, maps of prior episodes, or dataset-specific



Token Interface

The same action interface is used by the rule policy, learned policies, and AlpaSim transfer experiments.

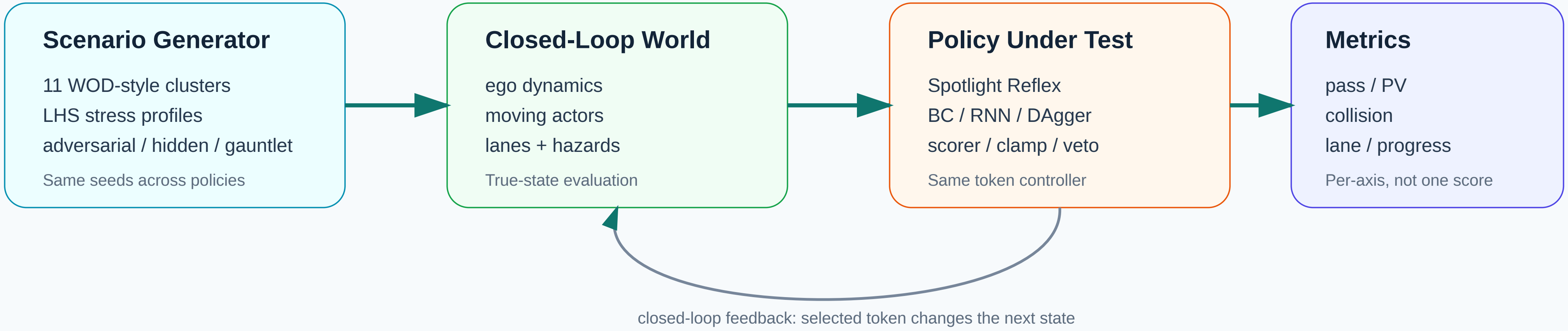
Token family	Examples	Role
longitudinal safety	stop, crawl, slow_yield	reduce collision pressure
nominal progress	maintain	continue along route
mild lateral evasion	nudge_left, nudge_right	create local clearance
emergency evasion	evasive_left, evasive_right	handle close hazards
recovery	lane_recover	return to lane center

This shared token interface makes failures diagnosable: we can separate bad candidate geometry from bad token selection.

Simulator Environment

Custom Simulator: Why It Exists

It is a controlled stress lab where state, proxy corruption, policy, and metrics are independently manipulable.



Proxy Perturbations

heading bias, actor latency
route offset, lane scale
feature noise

Question It Answers

Is AlpaSim failure just an adapter bug?
No: the same tradeoffs appear
under controlled proxy corruption.

Key Output

288 matched proxy cases
1440 agent rollouts
tradeoffs by metric axis

Demo: Minimal-Shot Success And Failure

<table>
<tr><th>Baseline failure</th><th>Geometry-grounded success</th></tr>
<tr><td>
</td><td>
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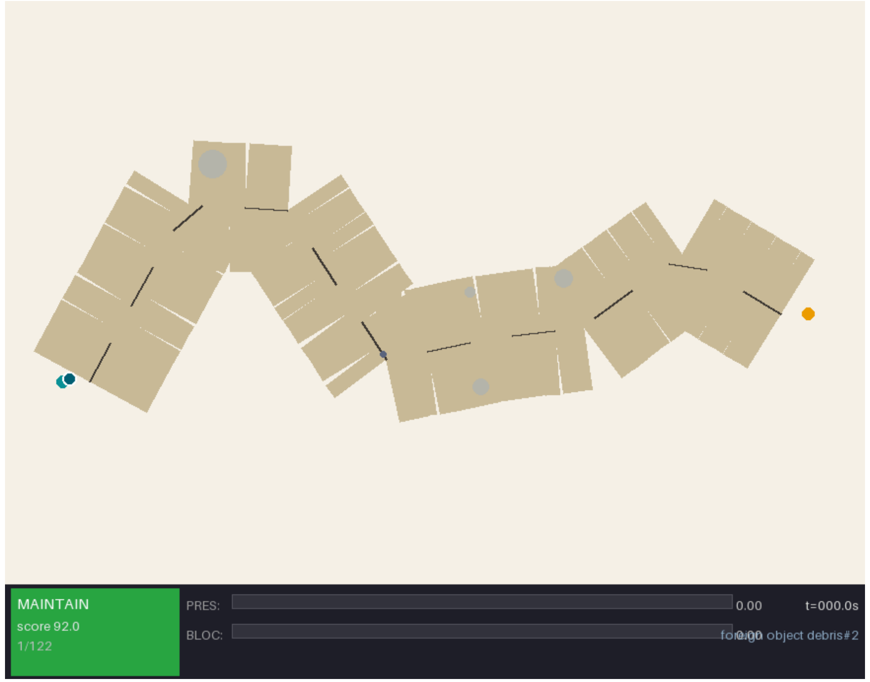
</table>

Same scenario family: wrong-way actor, low visibility, night.

The baseline extrapolates into the hazard. Spotlight selects an evasive token,

More Long-Tail Scenarios

<table>
<tr><th>Construction zone</th><th>Foreign object debris</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

The point is not photorealism. The point is controlled variation of geometry, hazards, clearances, and recovery choices.

Primary Simulation Result

Closed-loop simulation evidence package:

Suite	Runs	Collisions	Pass
WOD-style, all 11 clusters	110	0	100%
Compositional OOD	60	0	100%
Adversarial	60	0	100%
Hidden holdout	60	0	100%
Gauntlet	60	0	60%
Total	350	0	93.1%

COMPASS: **9.137 / 10** over 700 ranked runs.

Source: `README.md`, `artifacts/compass_evidence_report.json`

Baseline Comparison

Matched gauntlet comparison: same 420 scenarios, same seeds.

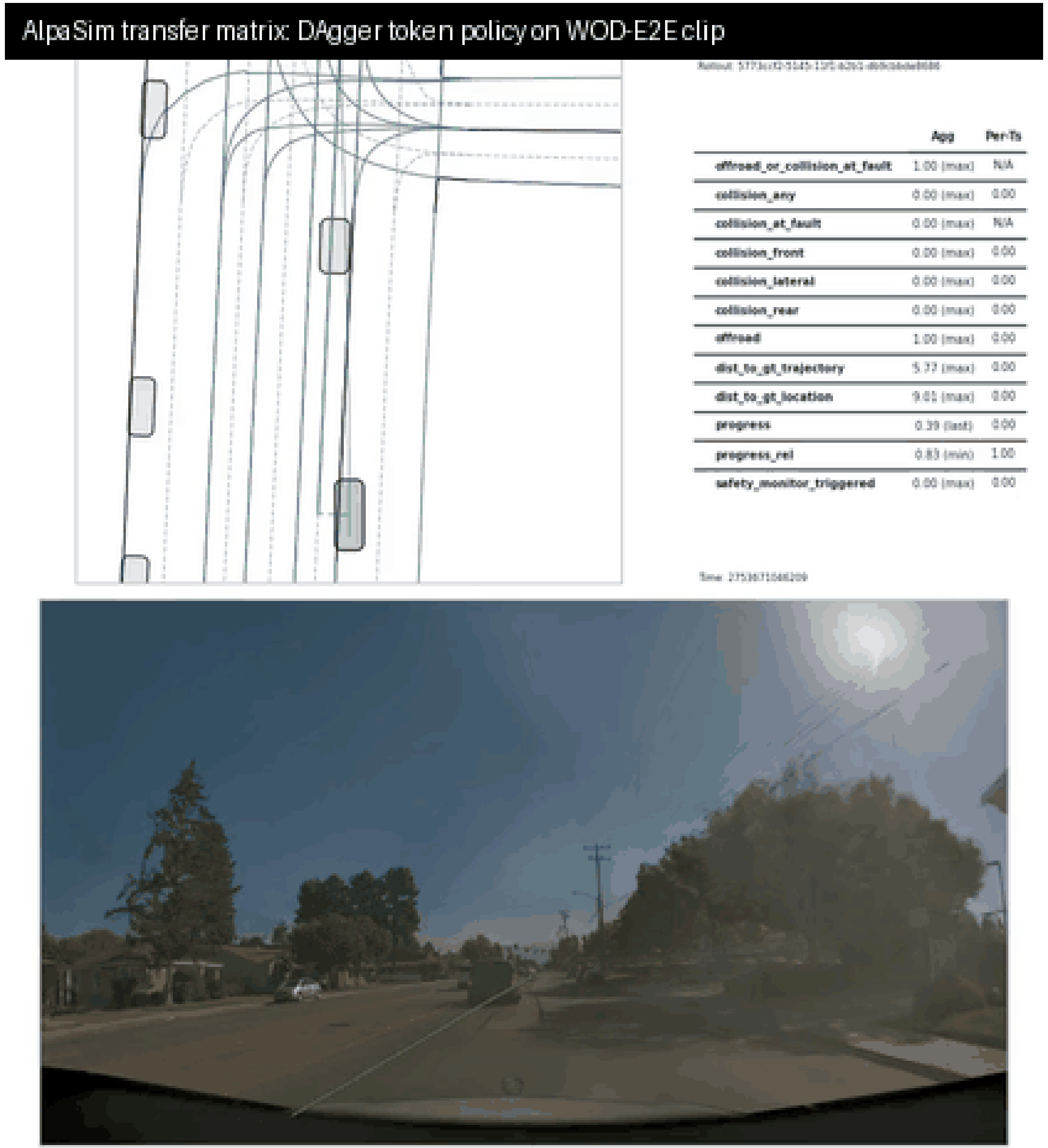
Policy	Pass rate	Collision rate
Baseline, no world-state reasoning	2.1%	20.5%
Spotlight Reflex	**57.6%**	**7.9%**

Takeaway for SoTA: explicit geometric reasoning gives a large improvement over a non-reasoning baseline in the randomized challenge environment.

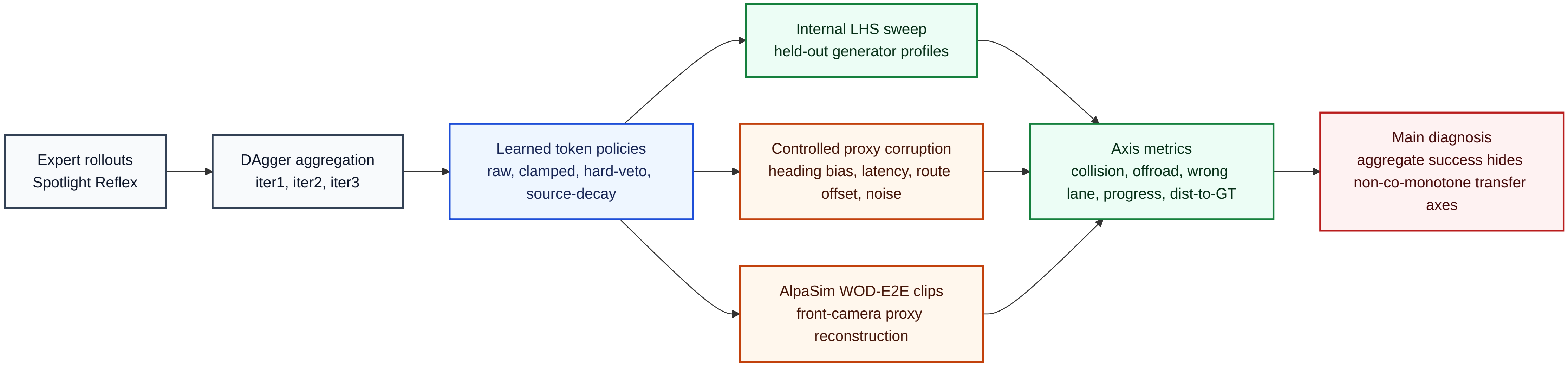
AlpaSim External Transfer

This is a WOD-E2E front-camera AlpaSim rollout with adapter map and metrics.

AlpaSim is not used to claim production safety. It is used to test whether the token interface survives a different observation and execution stack.



Transfer Diagnostic Schema



Mermaid-authored evidence pipeline: the same learned token policies are tested inside the simulator, under controlled proxy corruption, and in AlpaSim.

What Failed Externally

10 matched WOD-E2E clips completed by all four learned variants:

Model	Collision	Offroad	Wrong lane	Progress	Dist. m
Raw DAgger iter2	0.600	0.900	0.700	0.034	61.98
Clamped lateral	0.700	0.500	0.200	0.384	59.85
Hard-veto hybrid	0.800	0.200	0.700	0.837	166.06
Source decay	0.600	0.900	0.400	0.175	62.92

No single learned intervention fixed all axes. This is the main understood failure case: transfer breaks into collision, offroad, lane, and progress axes.

Source: `artifacts/alpasim_matrix10_analysis.md`.

WOD-E2E Auxiliary Track

479 WOD-E2E validation frames, segment-grouped 5-fold CV.

Selector	RFS	Folds	Role
Waymo baseline	7.022	-	reference
Local baseline	7.131	-	reference
Gate-only	7.803	5	selector baseline
RFF direct policy	7.834	5	learned baseline
GPU MLP + Cosmos 64d	**7.845**	5	best validation-CV
Oracle selector	9.264	5	upper bound

This is auxiliary evidence, not a hidden-test claim: the right trajectory often exists, but selection needs stronger grounded scene understanding.

Why The Project Is Novel

Judging axis	Evidence
technical excellence	simulator, token runtime, AlpaSim bridge, WOD-E2E harness
novelty	minimal-shot geometry-first token selection, per-axis transfer diagnosis
feasibility	small runtime, deterministic policy, clear deployment path
adherence to brief	custom simulation, randomized scenarios, videos, repo, audit trail

The strongest idea is not “one model wins everywhere.”

It is that minimal-shot autonomy should be evaluated by how failures decompose under new geometry, not only by aggregate pass rate.

What Funding Enables

The next prototype milestone is a real grounded-selection loop:

1. scale AlpaSim validation from 10 paired scenes to 100+ scenes
2. add camera-grounded hazard and lane reconstruction to reduce proxy mismatch
3. train an axis-aware selector that optimizes collision, lane, offroad, and progress separately
4. run the same system on off-road or closed-course RC vehicle footage
5. publish the simulator and audit protocol as a minimal-shot AV benchmark

The prize money would fund compute, WOD/AlpaSim storage, annotation checks, and
hardware for a small physical prototype.

Reproducibility

Start here:

```
README.md
presentation.md / presentation.pdf
docs/corl2027/paper.pdf
docs/corl2027/AUDIT.md
./scripts/run_corl2027_audit.sh
```

Submission evidence:

```
artifacts/final_submission_readiness_audit.json
artifacts/sota_judging_criteria_audit.json
artifacts/minimal_shot_claim_audit.json
artifacts/alpasim_matrix10_analysis.md
```

Branch:

Claim Boundary

Claim	Status
working minimal-shot AV simulator	yes
autonomous policy demo in randomized long-tail scenes	yes
WOD-E2E validation-CV selector result	yes, not hidden-test
AlpaSim external transfer diagnosis	yes, 10 paired shared clips
production AV safety claim	no
uniform positive learned method	no

The submission is strongest as a prototype plus diagnostic environment, with an

CoRL Research Appendix

The same repo also contains a CoRL 2027 diagnostic paper draft:

> Internal imitation-learning success can hide transfer-axis disagreement. We reproduce the mechanism under controlled proxy-state corruption and external
> AlpaSim transfer.

This appendix supports the SoTA submission but is not the main deck framing.

Internal Imitation-Learning Result

Held-out Latin-hypercube sweep: 12 profiles, seeds 1-10, Gauntlet / Adversarial / Hidden.

Agent	Overall pass / PV	Gauntlet	Adversarial	Hidden
Continuous-BC	77.69 / 3.52	77.50 / 4.31	75.42 / 2.08	83.33 / 1.67
Token-BC	76.85 / 4.26	77.64 / 4.86	71.67 / 2.92	82.50 / 3.33
Token-RNN-BC	78.43 / 2.96	79.17 / 2.92	74.58 / 2.50	81.67 / 4.17
Token-DAgger-BC, 2-	**94.54 / 0.19**	96.25 /	**90.83 / 0.00**	**91.67 /**

Controlled Proxy-State Test

The simulator reproduces AlpaSim-style tradeoffs when only policy-visible state is corrupted.

Perturbation	Raw p/c/l	Clamp	Hybrid	
clean	0.917/0.083/0.312	0.938/0.062/0.167	0.958/0.042/0.167	0.938/
actor latency	0.604/0.396/0.875	0.604/0.396/0.521	**0.854/0.146/0.208**	0.833/
route offset	0.938/0.062/0.229	0.875/0.125/0.021	0.896/0.104/0.062	0.938/
feature noise	0.917/0.083/0.917	0.938/0.062/0.146	0.917/0.083/0.125	0.896/

`p/c/l` means pass / collision / lane-violation rate.

Source: `artifacts/internal_proxy_transfer_medium.md`.

Grounding Context

Recent VLA work argues for physically grounded latent reasoning.

LaST-VLA uses 3D geometric priors and world-model dynamics to improve autonomous-driving VLA planning.

Our complementary lesson:

- grounding is useful
- but aggregate scores can hide metric-axis conflicts
- grounded selectors should report collision, lane, offroad, progress, and tracking separately

Reference: LaST-VLA, arXiv:2603.01928

Final Takeaway

The SoTA contribution is a complete minimal-shot AV prototype:

- custom randomized simulator
- auditable geometry-first policy
- scenario videos and external AlpaSim transfer
- WOD-E2E validation-CV selector harness
- transparent failure analysis

The research contribution is the diagnosis:

minimal-shot AV systems should be judged by grounded, per-axis transfer behavior,
not by offline accuracy or a single aggregate score.